

SEAT NO. _____

No. of Printed Pages: 5

[12]

SARDAR PATEL UNIVERSITY
T.Y.BCA (V - Semester) Examination

08/09/2021, Wednesday
10.00 AM TO 12.00 PM

US05FBCA01: Operation Research

Total Marks: 70

Q-1 Give answers of following Questions. (MCQs)

[10]

1. In graphical method the restriction on number of constraint is _____.
(a) 2 (b) 3
(c) not more than 3 (d) none of the above
2. The linear function of variables which is to be maximized or minimized is called _____.
(a) constraints (b) basic requirements
(c) objective function (d) none of them
3. Minimize $Z =$ _____.
(a) $-\text{maximize}(Z)$ (b) $\text{maximize}(-Z)$
(c) $-\text{maximize}(-Z)$ (d) none of the above
4. The coefficient of slack variable in the objective function is _____.
(a) $-M$ (b) $+M$
(c) 0 (d) none of them
5. The _____ variable is added to the constraint of less than equal to type.
(a) slack (b) Surplus
(c) artificial (d) basic
6. The _____ method's solution for transportation problem is sometimes an optimal solution itself.
(a) NWCM (b) VAM
(c) LCM (d) Row Minima
7. _____ Operation is carried out on a machine at a time.
(a) Two (b) only one
(c) at least one (d) none of them.
8. North – West corner refers to _____.
(a) top left corner (b) top right corner
(c) both of them (d) none of them
9. Activity which starts only after finishing other activity is called _____.
(a) dummy (b) Predecessor
(c) successor (d) none of them
10. _____ is indicated by dotted arrow.
(a) burst event (b) merge event
(c) dummy activity (d) none of them

Q-2 Fill in the blanks and True/False.

[08]

1. Operation Research is an art of winning a war without actually fighting it.(True/False)
2. LPP stands for _____.
3. Graphical Method is used for solving LPP having only Two Variable.(True/False)
4. The Solution available at the origin is termed as initial _____.
5. The North – West Corner method is made at Upper Left Corner of the Transportation table.(True/False)
6. Feasible solution is said to be _____ if it minimizes the total transportation cost.
7. Network Diagram is Pictorial representation of activities using arrows and nodes.(True/False)
8. An Event which represent the join completion of more than one activity is called _____.

Q-3 Give answers of following Questions.(Any Ten)

[20]

1. Define LPP in the mathematical form.
2. Define Operation research.
3. Define i] solution ii] Basic solution
4. Define slack variables.
5. Write the standard form of LPP for the following LPP:
$$\text{Maximize } Z = 13x_1 + 25x_2$$
$$\text{Subject to } 21x_1 + 3x_2 \leq 40, \quad 5x_1 + 2x_2 \leq 7, \quad x_1, x_2 \geq 0$$
6. What is the condition for optimality in simplex table?
7. What is transportation problem?
8. Write mathematical form of transportation problem.
9. Give mathematical form of assignment problem.
10. State any two rules for drawing network diagram.
11. Define: Total Float & Free Float
12. What is Dynamic Programming Problem?

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Q-4 Long Questions.(Attempt any Four).

[32]

- 1 A farmer is deciding for breeding of two crops. The first crop (C1) has 36 units of nutrient 1, 30 units of nutrient 2 and 20 units of nutrient 3. The second crop (C2) has 16 units of nutrient 1, 12 units of nutrient 2 and 10 units of nutrient 3. Daily requirement of each nutrient 1, 2 and 3 is 108, 96 and 100 units respectively. Each unit of first crop costs Rs. 20 and second crop costs Rs. 40. Formulate the LPP such that cost is minimum.

- 2 Find a solution for the following LP problem using graphical method.

$$\text{Maximize } Z = 3x_1 + 9x_2$$

$$\text{Subject to } x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$\text{And } x_1, x_2 \geq 0$$

- 3 Solve the below LPP using simplex method.

$$\text{Max } Z = 3x_1 + 2x_2$$

$$\text{Subject to } 2x_1 + x_2 \leq 5,$$

$$x_1 + x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

- 4 Solve the below LPP using Big-M method.

$$\text{Min } Z = 3x_1 + 8x_2$$

$$\text{Subject to } x_1 + x_2 = 200,$$

$$x_1 \leq 80,$$

$$x_2 \geq 60$$

$$x_1, x_2 \geq 0$$

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- 5 Obtain the optimal solution to Transportation Problem using northwest corner method.

	Destination			
	A	B	C	Supply
F ₁	4	5	1	40
F ₂	3	4	3	60
F ₃	6	2	8	70
Demand	70	40	60	

- 6 Obtain the initial solution to above TP using least cost method.

	Destination				
	A	B	C	D	Supply
F ₁	2	3	11	7	6
F ₂	1	0	6	1	1
F ₃	5	8	15	9	10
Demand	7	5	3	2	

- 7 A project has the following time schedule :

Activity	Time In month	Activity	Time In month
1-2	6	3-5	8
1-3	9	2-6	12
2-4	3	4-6	7
3-4	4	5-6	1

Construct PERT network and compute total float for each activity.
Find Critical path with its duration.

(4)

- 8 In a machine shop 5 different products are being manufactured each requiring time on two different machines A and B are given in the table below:

Product	1	2	3	4	5
Machine-A	5	1	9	3	10
Machine B	2	6	7	8	4

Find an optimal sequence of processing of different product in order to minimize the total manufactured time for all product. Find total ideal time for two machines and elapsed time.



